Azure Sentinel Honeypot Analysis

Reporting Period

2 Feb - 8 Feb

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Introduction

In an era of automated exploitation, understanding both the “how” and “who” behind an attack is essential for resilient defence. This report summarises intelligence gathered from a controlled honeypot environment during 2–8 February 2026, focusing on unusual authentication patterns and scanning activity directed at core protocols.

By correlating the top 10 malicious IPs with relevant CVE severities, we identified the primary techniques used by persistent adversaries. These insights inform the remediation steps required to strengthen organisational security.

Scope

This analysis highlights targeted usernames and CVE exploitation attempts. Specific password patterns were not examined. The firewall was intentionally disabled to observe attacker behaviour, which resulted in multiple brute-force attempts — reinforcing the need for strong credential-hardening measures.

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Methodology

1. Environment Setup
Platform: Microsoft Azure
Instance Type: Ubuntu 24.04 LTS -x64

2. Honeypot Creation
Step 1a: Provisioning the Azure Virtual Machine

- VM name: CS-Payroll-Backup
- Resource Group: Soc-Lab-Test-RG
- Region: Central US
- Workspace: Log Analytics Workspace.

b. Configured the NSG to allow inbound traffic from all protocol, source, destination and ports (SSH was allowed and restricted to my IP for remote access).
c. Deployed the Azure Monitor Linux Agent (AMLA) and created Data Collection Rules (DCR) to forward auth.log and syslog events directly into the Log Analytics Workspace.
d. Integrated a Watchlist in Sentinel to flag and map malicious IP addresses to geographical origins.

Step 2. Connecting the remote desktop to the VM

a. To update the system, install the desktop environment and set up a connection between the xrdp and the VM:

```
sudo apt update && sudo apt upgrade -y
sudo apt install xfce xfce4-goodies -y
sudo apt install xrdp -y
sudo systemctl enable xrdp
echo xfce4-session > ~/.xsession
```

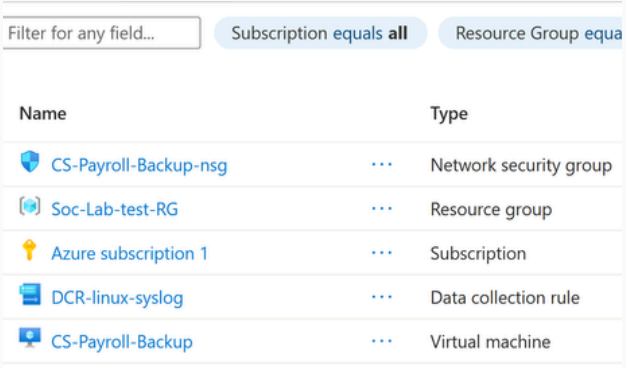
b. Add the xrdp user to the ssl-cert group:

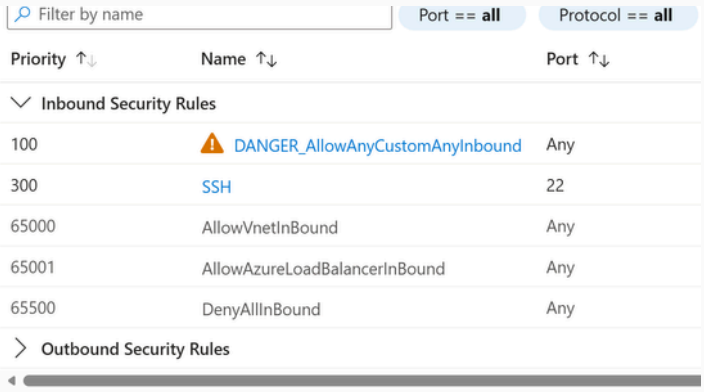
```
sudo adduser xrdp ssl-cert
```

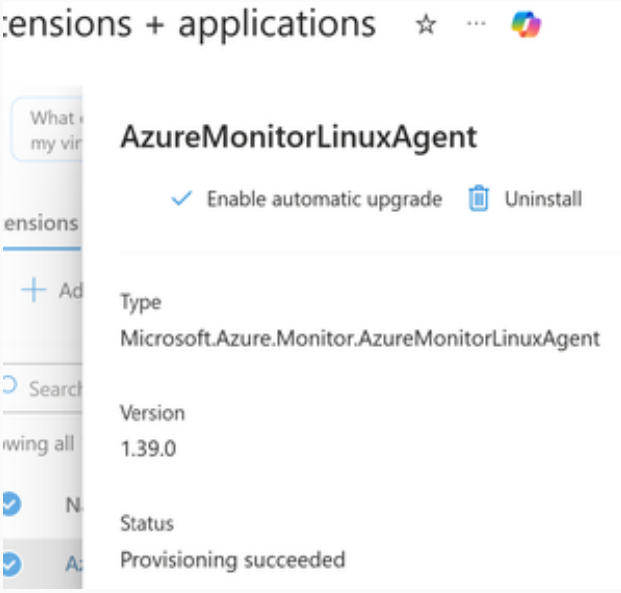
c. restart the xrdp:

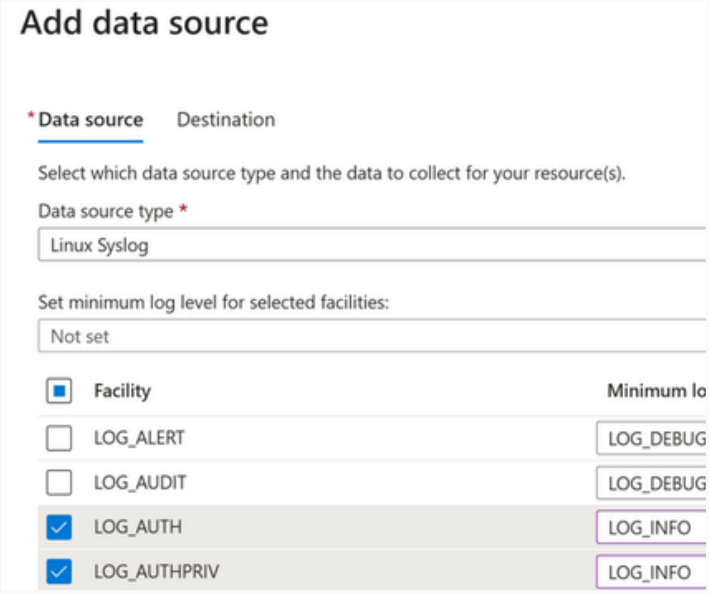
```
sudo systemctl restart xrdp
```

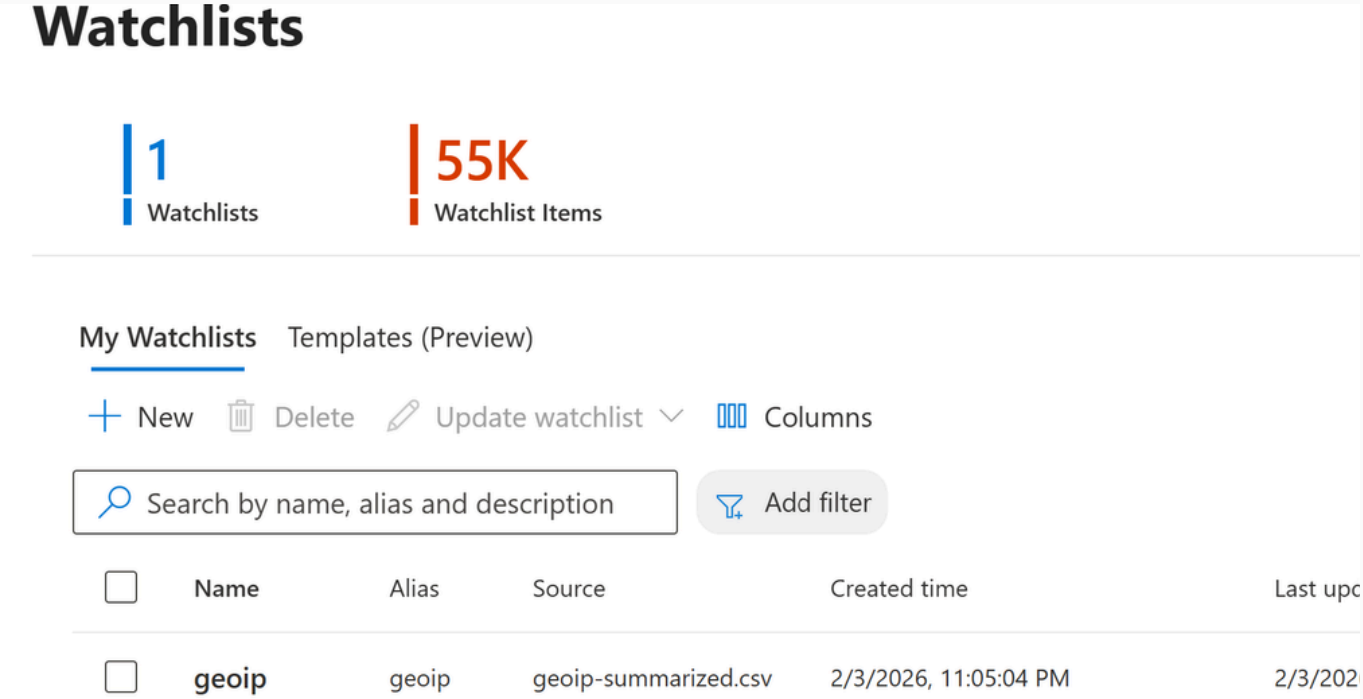
Screenshots for Methodology











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FEB 2 - FEB 8

Global Threat Map

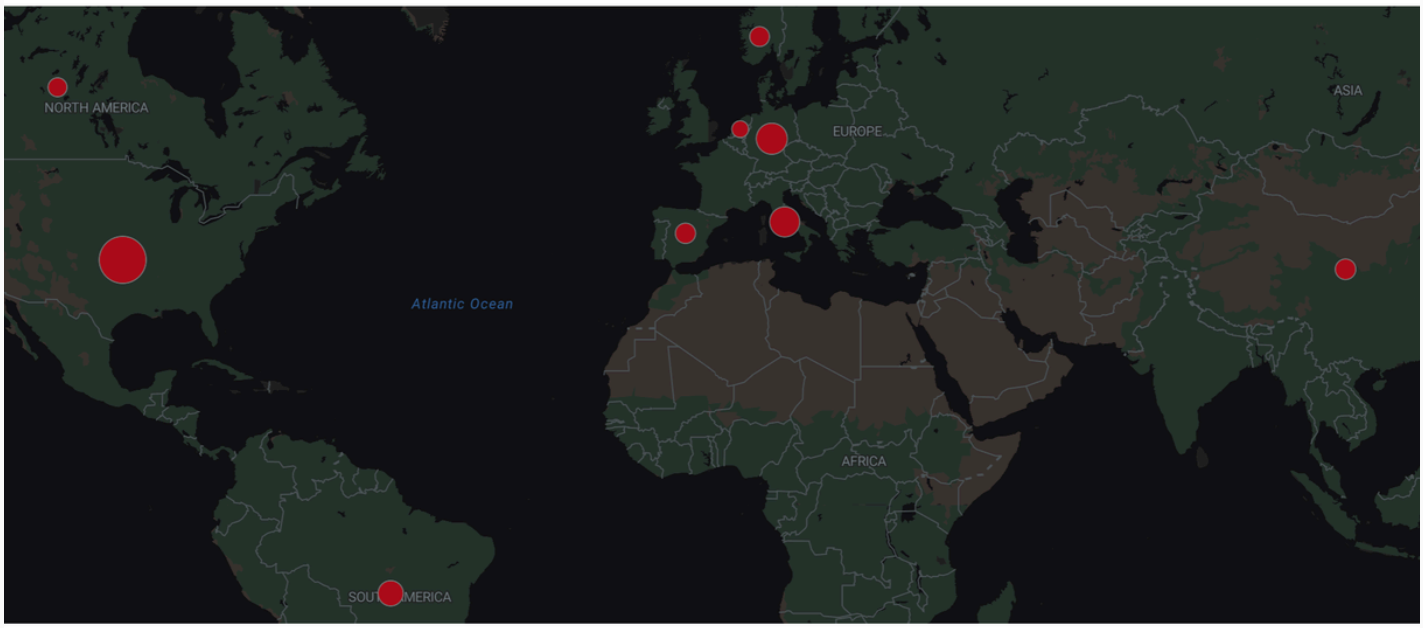

Fig 1. Geographic origin of Attackers' IP

Total Attacks
2,461

Most used attack
Brute-Force

Attacks Targeting SSH
2307

The persistent nature of automated attacks is evident in the Global Threat Map above.

During the reporting period 2,461 attacks were captured, originating from diverse global locations, forming several high-intensity clusters targeted at the Honeypot.

These results reinforce the need for robust security controls in an unfiltered network environment.

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FEB 2 - FEB 8

Attack Timing and Origin

Active Times (UTC)

Tuesday 10 PM - 11 PM

Wednesday 12 AM - 1 AM

Saturday 6 PM - 11PM

Top Locations

U.S.A 54%	South Korea 17%	Germany 12%
Italy 10%	Brazil 4%	

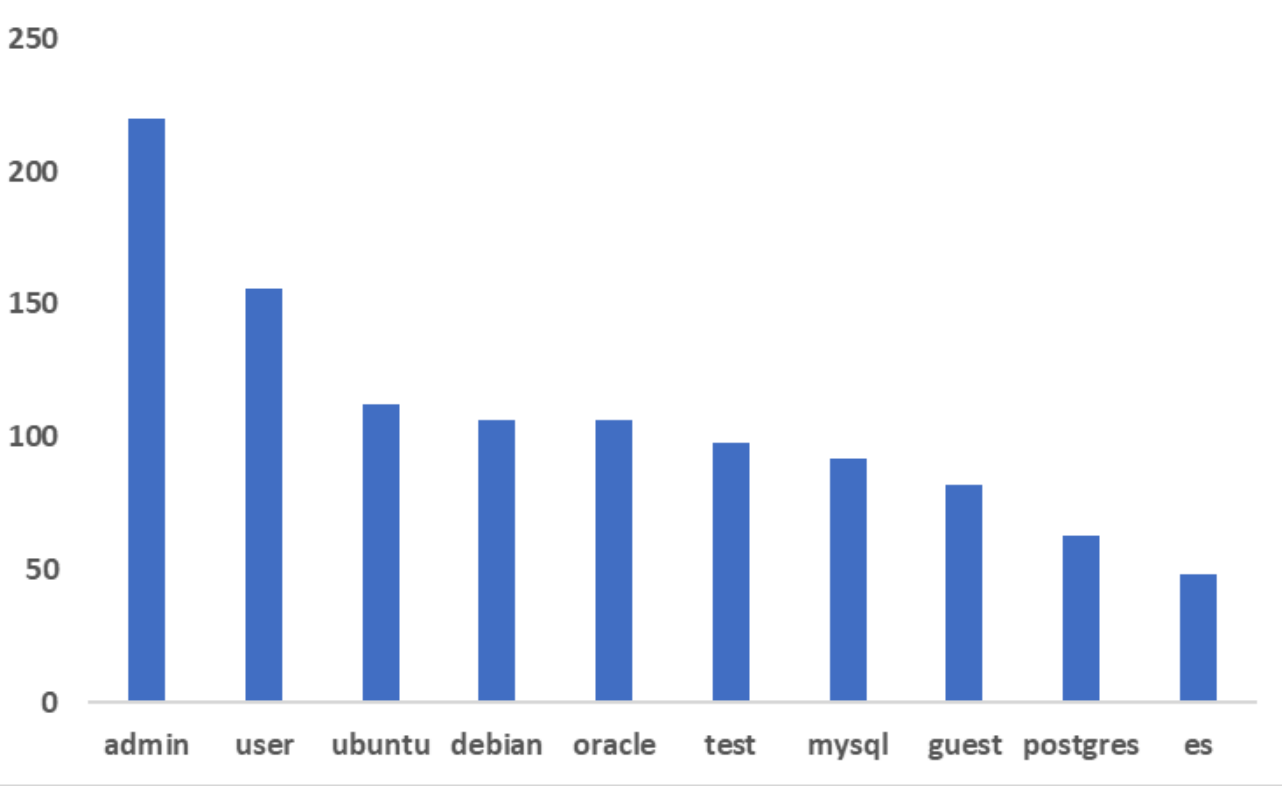
During the reporting period, activities peaked on Tuesday, Wednesday, and Saturday, with sustained activity across the day. The most concentrated surge took place from 6 PM to 11 PM, accounting for 54.8% of total attacks.

In terms of attack origin, the United States accounted for 54% of all activity, likely influenced by the honeypot's Central US hosting location. South Korea followed with 17%, while Germany, Italy and Brazil all recorded 12%, 10%, and 4% respectively.

These lower figures may reflect regional differences in scanning intensity and the concentration of botnet infrastructure. Overall, the distribution highlights the global and continuous nature of automated probing for exploitable systems.

Attacker Activity Analysis

Most Targeted Usernames



The chart highlights the ten most frequently targeted usernames and reveals a clear pattern: attackers are relying on automated tools that systematically attempt default, common, and high-privilege accounts rather than making random guesses.

- The most targeted username, “admin” with 220 attempts, reflects a direct effort to gain full system control.
- Usernames such as “Ubuntu,” “Debian,” and “Oracle” (each exceeding 100 attempts) indicate probing for default operating-system accounts that are often left unchanged after installation
- The appearance of “mysql” and “postgres” shows automated attempts to identify exposed database services that could be leveraged to access sensitive information.
- Entries like “user,” “test,” and “guest” represent attempts to exploit weak, temporary, or poorly maintained accounts that commonly exist in misconfigured environments.

Top 10 IPs Matched to CVE Attack attempts

Source IP	Attempt Count	CVE ID	Vulnerability name	Severity
115.190.43.239	1072	CVE-2024-6387; CVE-2018-15473	regreSSHion; User enumeration	High; Medium
134.209.203.247	914	CVE-2024-6387; CVE-2018-15473	regreSSHion; User enumeration	High; Medium
159.65.193.188	804	CVE-2024-6387; CVE-2018-15473	regreSSHion; User enumeration	High; Medium
209.38.38.12	700	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High
45.156.87.246	674	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High
94.26.106.110	674	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High
167.71.68.179	668	CVE-2024-6387; CVE-2018-15473	regreSSHion; User enumeration	High; Medium
164.92.155.97	275	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High
2.57.122.210	66	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High
221.182.17.158	59	CVE-2018-15473; CVE-2024-6387	User enumeration; regreSSHion	Medium; High

Recommendations for SOC Teams

Based on the comprehensive threat intelligence gathered, the following mitigation strategies are recommended to strengthen the organisation's infrastructure against the identified attack vectors.

Vulnerability Management

- Prioritise remediation of high-severity CVEs to minimise immediate risk.
- Implement SIEM detection rules to flag exploitation attempts, brute-force activities and unusual authentication patterns.
- Audit and close all unused or unnecessary ports to reduce the attack surface.

Intelligence-Driven IP Filtering

- Correlate attacker IPs with threat intelligence sources to identify associations with malicious infrastructure.
- Integrate identified attacker IPs into firewall, WAF or SIEM blocklists.
- Implement logic to detect scanning behaviour, such as rapid connection attempts on multiple ports.

Credential Hardening

- Enforce the use of complex passwords, MFAs and password rotation.
- Conduct periodic access control and credential audits.
- Disable unused default or legacy accounts to prevent unauthorised access.

Continuous log Monitoring

- Monitor for unusual network traffic that may indicate scanning, lateral movement, or data exfiltration.
- Ensure sensitive services such as database ports are not exposed to the public internet.
- Maintain continuous log visibility across systems for prompt detection of anomalies.

Conclusion

The data captured during this honeypot session highlights a persistent and evolving threat landscape that demands a proactive, intelligence-led defence strategy. Implementing the recommended remediation measures will help the organisation disrupt the brute-force behaviour identified during analysis. Sustained resilience requires continuous integration of high-quality threat intelligence into the SIEM, so defensive logic evolves as quickly as automated attackers. As adversary techniques become more sophisticated, the SOC should prioritise automating protective controls to safeguard the infrastructure over the long term.